Dataset: California Housing (source: StatLib, 1990 census) — columns: longitude, latitude, housing_median_age, total_rooms, total_bedrooms, population, households, median_income,median_house_value, ocean_proximity

Questions:

1. Create an income_cat column by binning median_income with edges [0, 1.5, 3.0, 4.5, 6.0, np.inf] labeled 1–5. Using StratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42), split the data. Report the count of each income_cat value in both the train and test sets.

2. Build a ColumnTransformer with: numeric pipeline = SimpleImputer(strategy="median") → StandardScaler(); categorical pipeline = OneHotEncoder(handle_unknown="ignore") on ocean_proximity. Fit-transform the training set and report the shape of the resulting array.

3. Using this pipeline, train LinearRegression(), DecisionTreeRegressor(random_state=42), and RandomForestRegressor(n_estimators=100, random_state=42). Using cross_val_score with cv=5 and scoring="neg_root_mean_squared_error", report the mean ± std RMSE for each model.

4. Run GridSearchCV on RandomForestRegressor over n_estimators $\in$ {50, 100, 200} and max_features $\in$ {2, 4, 6, 8} with cv=5. Report the best hyperparameter combination and its CV RMSE.

5. Evaluate the best tuned model on the held-out test set. Report Test RMSE, MAE, and R^2.

6. Using only NumPy, implement linear regression via the Normal Equation ($\theta = (X^TX)^{-1}X^Ty$) with median_income and housing_median_age (plus a bias/intercept term) as the only two features to predict median_house_value. Report your computed θ values and their % difference from sklearn.linear_model.LinearRegression fit on the same two features.

7. Plot a geographic scatter plot (longitude vs latitude, point size $\propto$ population, color $\propto$ median_house_value, alpha=0.4).

8. Plot a histogram of median_house_value.

9. Plot predicted vs. actual median_house_value on the test set (scatter plot with a red dashed diagonal reference line).

10. Plot the top-10 feature importances from the tuned Random Forest as a horizontal bar chart.